

O Princípio da Homogeneidade e Análise Dimensional

Professor Edson Mesquita, Engenharia Naval e Oceânica, D. Sc.

Homogeneidade dimensional

Dimensions Associated with Common Physical Quantities					
	<i>FLT</i> System	<i>MLT</i> System		<i>FLT</i> System	<i>MLT</i> System
Acceleration	LT^{-2}	LT^{-2}	Momentum	FT	MLT^{-1}
Angle	$F^0L^0T^0$	$M^0L^0T^0$	Power	FLT^{-1}	ML^2T^{-3}
Angular acceleration	T^{-2}	T^{-2}	Pressure	FL^{-2}	$ML^{-1}T^{-2}$
Angular velocity	T^{-1}	T^{-1}	Specific heat	$L^2T^{-2}\Theta^{-1}$	$L^2T^{-2}\Theta^{-1}$
Area	L^2	L^2	Specific weight	FL^{-3}	$ML^{-2}T^{-2}$
Density	$FL^{-4}T^2$	ML^{-3}	Strain	$F^0L^0T^0$	$M^0L^0T^0$
Energy	FL	ML^2T^{-2}	Stress	FL^{-2}	$ML^{-1}T^{-2}$
Force	F	MLT^{-2}	Surface tension	FL^{-1}	MT^{-2}
Frequency	T^{-1}	T^{-1}	Temperature	Θ	Θ
Heat	FL	ML^2T^{-2}	Time	T	T
Length	L	L	Torque	FL	ML^2T^{-2}
Mass	$FL^{-1}T^2$	M	Velocity	LT^{-1}	LT^{-1}
Modulus of elasticity	FL^{-2}	$ML^{-1}T^{-2}$	Viscosity (dynamic)	$FL^{-2}T$	$ML^{-1}T^{-1}$
Moment of a force	FL	ML^2T^{-2}	Viscosity (kinematic)	L^2T^{-1}	L^2T^{-1}
Moment of inertia (area)	L^4	L^4	Volume	L^3	L^3
Moment of inertia (mass)	FLT^{-2}	ML^2	Work	FL	ML^2T^{-2}

Variáveis, parâmetros ou números adimensionais

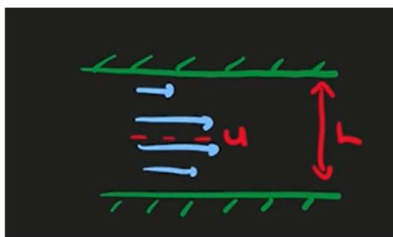
$$\rho V_0^2 L^2$$

$$V_0$$

$$L$$

$$V_0 = u$$

$$L = b$$



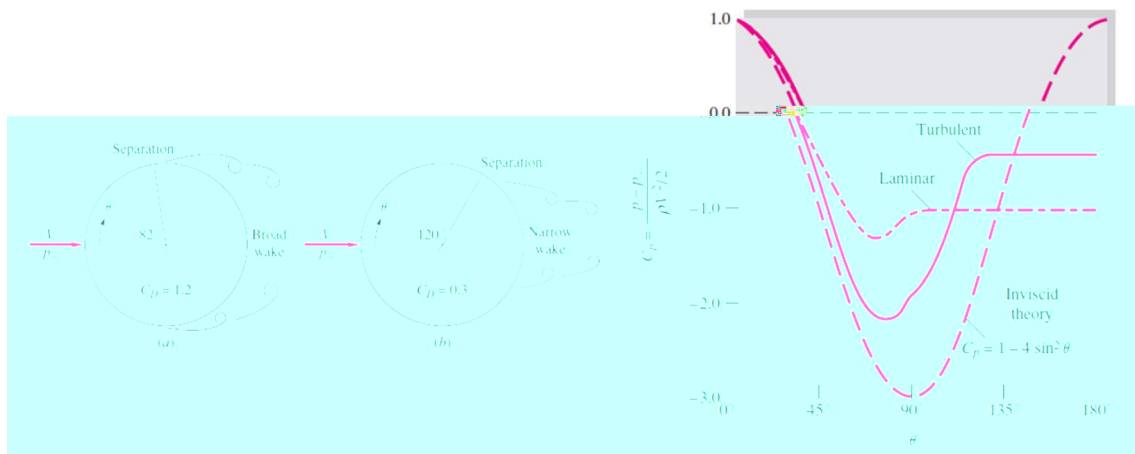
$$K_e$$

$$K_e \cong mu^2$$

$$K_e = F \cdot L$$

$$F_{inércia} = \frac{mu^2}{L} = \frac{\rho u^2 L^3}{L} = \rho u^2 L^2$$

O número de Euler



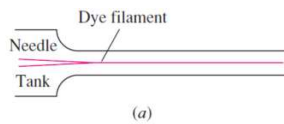
O número de Euler

$$\Delta p = \frac{F}{A} = \frac{F}{L^2} \rightarrow F_{\text{pressão}} = \Delta p L^2$$

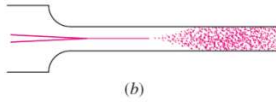
$$\frac{F_{\text{pressão}}}{F_{\text{inércia}}} = \frac{\Delta p L^2}{\rho V_0^2 L^2} = \frac{\Delta p}{\rho V_0^2}$$

Eu

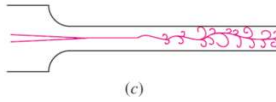
$$Eu = \frac{\Delta p}{\rho V_0^2}$$



(a) Regime laminar.



(b) Regime turbulento.



(c) Regime turbulento, foto instantânea.

$(Re_D)_{cr} \cong 2,3 \times 10^3$ para dutos comerciais de diâmetro D .

$$\tau = \frac{F_{viscosa}}{A} \rightarrow F_{viscosa} = \tau A$$

$$\tau = \mu \frac{\Delta u}{\Delta y}$$

$$F_{viscosa} = \left(\mu \frac{\Delta u}{\Delta y} \right) A$$

$$\Delta u \rightarrow u \quad \Delta y \rightarrow L \quad A \rightarrow L^2$$

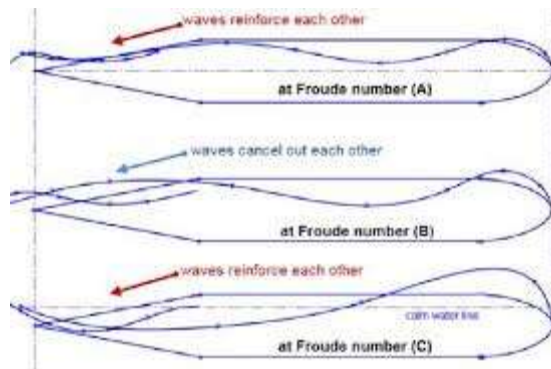
$$F_{viscosa} = \mu \frac{uL^2}{L}$$

$$\frac{F_{inércia}}{F_{viscosa}} = \frac{\rho u^2 L^2}{\mu \frac{V_0 L^2}{L}} = \frac{\rho u L}{\mu}$$

$$Re$$

$$Re = \frac{\rho u L}{\mu}$$

O número de Froude



$$F_{gravidade} = mg = \rho \cdot Vol \cdot g$$

$$F_{gravidade} = \rho L^3 g$$

$$\frac{F_{inércia}}{F_{gravidade}} = \frac{\rho V_0^2 L^2}{\rho L^3 g} = \frac{V_0^2}{gL}$$

Fr

$$Fr = \sqrt{\frac{V_0^2}{gL}} = \frac{V_0}{\sqrt{gL}}$$

O número de Weber

$$F_{superf} = \sigma L$$

σ

$$\frac{F_{inércia}}{F_{superf}} = \frac{\rho u^2 L^2}{\sigma L} = \frac{\rho u^2 L}{\sigma}$$

$$Wn = \frac{\rho u^2 L}{\sigma}$$

Análise dimensional

-
-
-

O teorema de Buckingham (1914)

$$u \qquad \qquad \qquad A \qquad \qquad \qquad Q = uA$$

$$p,u,\rho,L$$

$$p\colon [ML^{-1}T^{-2}], u\colon [LT^{-1}], \rho\colon [ML^{-3}], L\colon [L]$$

$$\Delta p,u,\rho,L$$

$$u,\rho,L$$

$$\pi_1=\Delta p u^a \rho^b L^c$$

$$a,b\in c$$

$$[M^0L^0T^0]=[ML^{-1}T^{-2}]\,[LT^{-1}]^a[ML^{-3}]^b[L]^c$$

$$[M^0L^0T^0]=[M]^{1+b}[L]^{-1+a-3b+c}[T]^{-2-a}$$

$$M\colon 0=1+b\qquad\rightarrow\qquad b=-1$$

$$L\colon 0=-1+a-3b+c$$

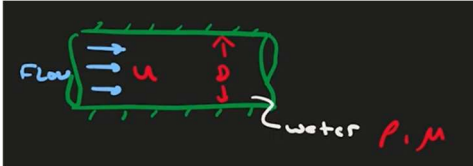
$$T\colon 0=-2-a\qquad\rightarrow\qquad a=-2$$

$$c=0$$

$$\pi_1=\Delta p u^{-2} \rho^{-1} L^0$$

$$\pi_1 \qquad \qquad \qquad Eu$$

$$\pi_1 = Eu = \frac{\Delta p}{\rho V_0^2}$$



$$u, D, \mu, \rho$$

$$u: [LT^{-1}], D: [L], \mu: [ML^{-1}T^{-1}] \rho: [ML^{-3}]$$

$$u, \mu, \rho$$

$$\pi_1 = D \rho^a \mu^b u^c$$

$$a, b \text{ e } c$$

$$[M^0 L^0 T^0] = [L] [ML^{-3}]^a [ML^{-1}T^{-1}]^b [LT^{-1}]^c$$

$$[M^0 L^0 T^0] = [M]^{a+b} [L]^{1-3a-b+c} [T]^{-b-c}$$

$$M: 0 = a + b \quad \rightarrow \quad a = -b$$

$$L: 0 = 1 - 3a - b + c$$

$$T: 0 = -b - c \quad \rightarrow \quad c = -b$$

$$c = 0$$

$$L: 0 = 1 - 3a - b + c \quad a = -b \quad c = -b \quad \rightarrow 1 + 3b - b - b = 0 \rightarrow b = -1$$

$$\text{Então } a = 1 \text{ e } c = 1$$

$$\pi_1 = D \rho^1 \mu^{-1} u^1$$

$$\pi_1$$

$$Re$$

$$\pi_1 = Re = \frac{\rho u D}{\mu}$$

Os coeficientes de Resistência de Forma, Resistência Friccional e Resistência de Ondas na hidrodinâmica do navio

Determinação do π_1

$$\pi_1 = R_{forma} \rho^a L^b U^c$$

$$[M^0 L^0 T^0] = [M^1 L^1 T^{-2}] [M^1 L^{-3}]^a [L^1]^b [L^1 T^{-1}]^c$$

$$0 = M^{1+a} L^{1-3a+b+c} T^{-2-c}$$

$$\pi_1 = R_{forma} \rho^{-1} L^{-2} U^{-2}$$

$$\pi_1 = \frac{R_{forma}}{\rho L^2 U^2}$$

Determinação do π_2

$$\pi_2 = \mu \rho^a L^b U^c$$

$$[M^0 L^0 T^0] = [M^1 L^{-1} T^{-1}] [M^1 L^{-3}]^a [L^1]^b [L^1 T^{-1}]^c$$

$$0 = M^{1+a} L^{-1-3a+b+c} T^{-1-c}$$

$$\pi_2 = \mu \rho^{-1} L^{-1} U^{-1}$$

$$\pi_2 = \frac{\mu}{\rho L U}$$

$$\pi_2 = R n^{-1}$$

$$\pi_1 \quad \pi_2$$

Determinação do π_3

$$\pi_3 = g \rho^a L^b U^c$$

$$[M^0L^0T^0] = [L^1T^{-2}][M^1L^{-3}]^a[L^1]^b[L^1T^{-1}]^c$$

$$0 = M^aL^{1-3a+b+c}T^{-1-c}$$

$$\pi_3 = g\rho^0L^1U^{-2}$$

$$\pi_3 = \frac{gL}{U^2}$$

$$\pi_3 = Fn^{-2}$$

Determinação do π_4

$$\pi_4 = \sigma\rho^aL^bU^c$$

$$[M^0L^0T^0] = [MT^{-2}][M^1L^{-3}]^a[L^1]^b[L^1T^{-1}]^c$$

$$0 = M^{1+a}L^{-3a+b+c}T^{-2-c}$$

$$\pi_4 = \sigma\rho^{-1}L^{-1}U^{-2}$$

$$\pi_4 = \frac{\sigma}{\rho LU^2}$$

$$\pi_4 = Wn^{-1}$$

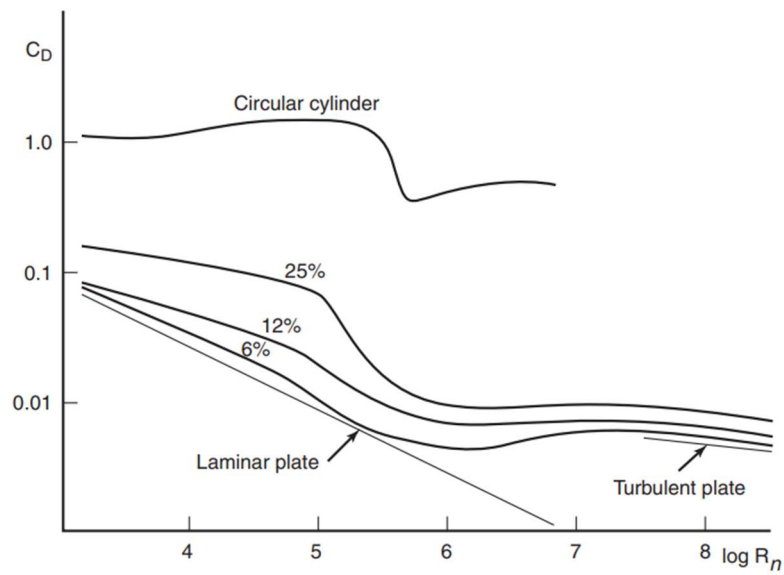
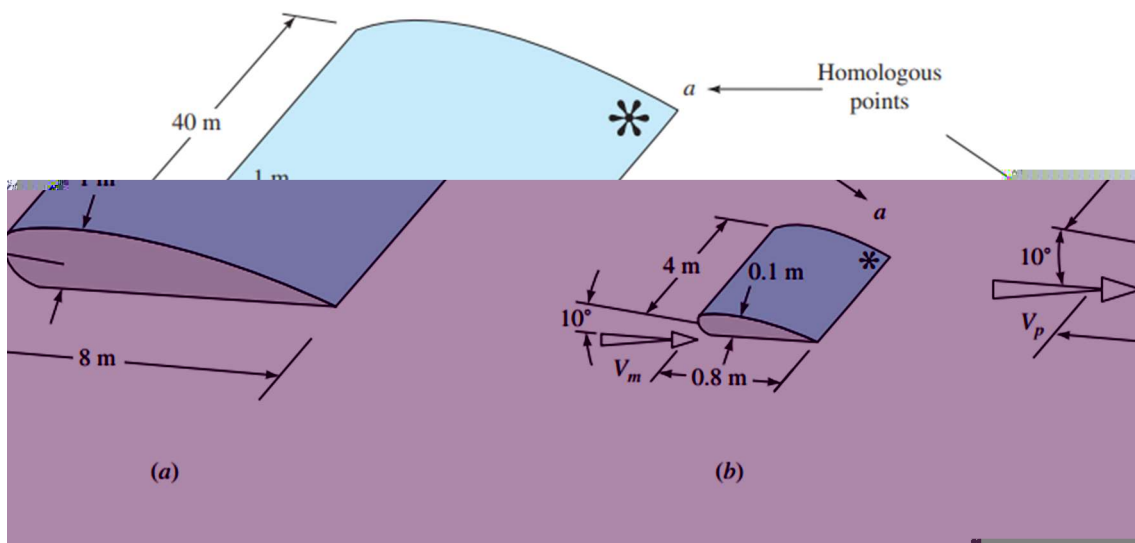
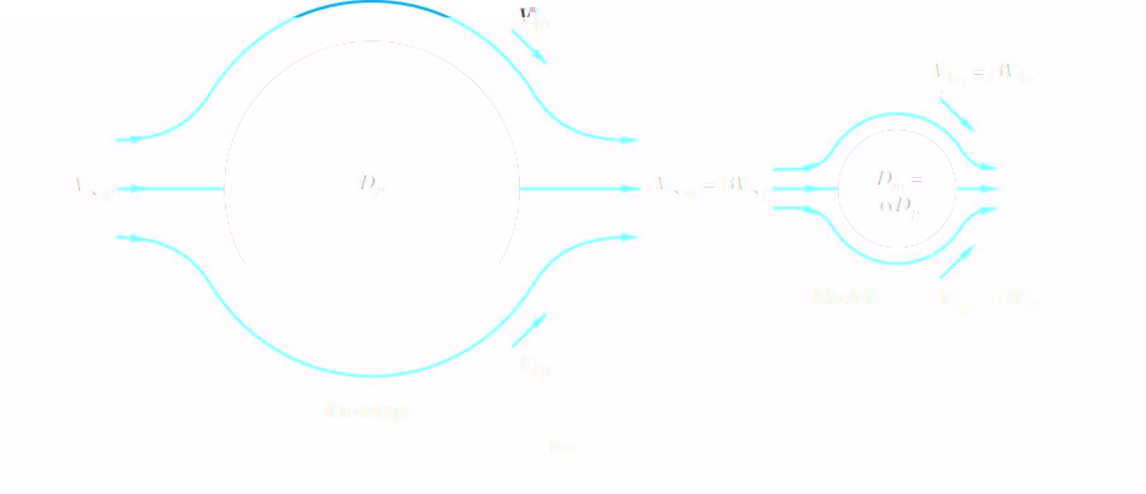


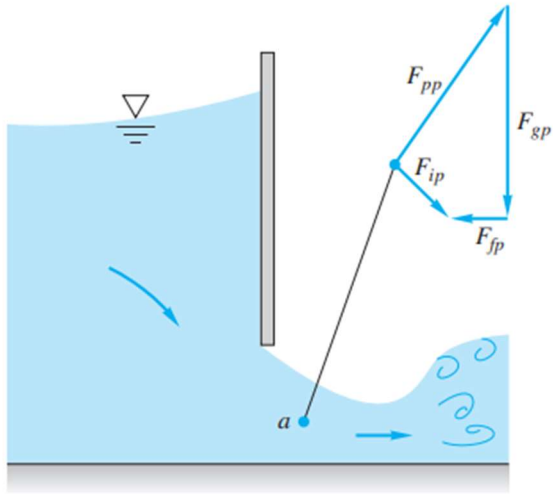
Figure 7.18 Reynolds number dependence of section drag.

O conceito da similaridade

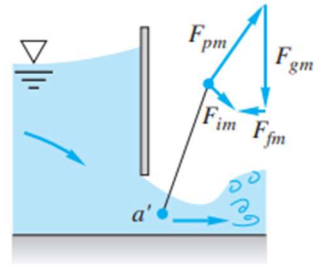
1. The model nose radius must be one-tenth as large.
2. The model surface roughness must be one-tenth as large.







(a)



(b)

O dilema da similaridade em testes com modelos

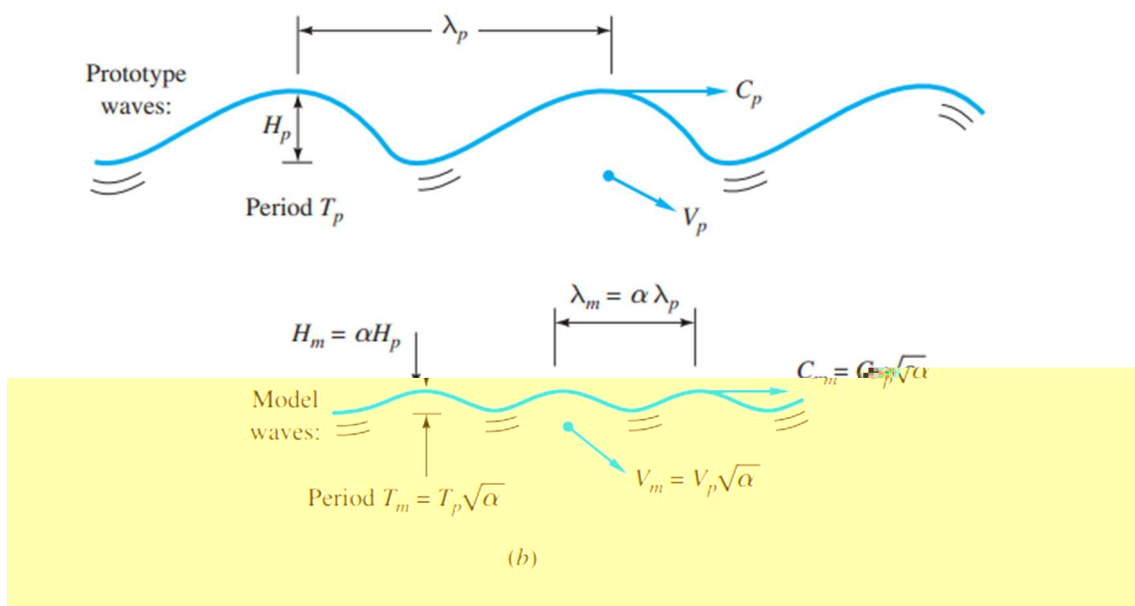
$$Fr_m = \frac{V_m^2}{gL_m} = \frac{V_p^2}{gL_p} = Fr_p$$

α

$$L_m = L_p \alpha$$

$$\frac{V_m^2}{V_p^2} = \frac{L_m}{L_p} \rightarrow \frac{V_m}{V_p} = \sqrt{\frac{L_m}{L_p}} = \sqrt{\alpha}$$

$$\frac{T_m}{T_p} = \frac{L_m/V_m}{L_p/V_p} = \frac{\alpha}{\sqrt{\alpha}} = \sqrt{\alpha}$$



$$\frac{V_m L_M}{v_m} = \frac{V_p L_p}{v_p}$$

$$\frac{v_m}{v_p} = \frac{L_M V_m}{L_p V_p} = \alpha \sqrt{\alpha} = \alpha^{3/2}$$

$$\alpha = 0.1 \quad \alpha^{3/2} = 0.032$$

Fluid	μ , kg/(m · s) [†]	Ratio $\mu/\mu(\text{H}_2)$	ρ , kg/m ³	ν , m ² /s [†]	Ratio $\nu/\nu(\text{Hg})$
Hydrogen	9.0 E-6	1.0	0.084	1.05 E-4	910
Air	1.8 E-5	2.1	1.20	1.50 E-5	130
Gasoline	2.9 E-4	33	680	4.22 E-7	3.7
Water	1.0 E-3	114	998	1.01 E-6	8.7
Ethyl alcohol	1.2 E-3	135	789	1.52 E-6	13
Mercury	1.5 E-3	170	13,550	1.16 E-7	1.0
SAE 30 oil	0.29	33,000	891	3.25 E-4	2,850
Glycerin	1.5	170,000	1,260	1.18 E-3	10,300

[†]1 kg/(m · s) = 0.0209 slug/(ft · s); 1 m²/s = 10.76 ft²/s.